The Challenge of Analyzing a Dynamic Text:
Why the Voynich Manuscript Resists Systematic Analysis

Torsten Timm

Abstract

The Voynich Manuscript (MS 408) has resisted all attempts at systematic analysis for over a century. This paper argues that the persistent failure of cryptographic, linguistic, and statistical approaches stems from a shared foundational assumption: that the manuscript's text was produced by a static system—whether a fixed cipher, a natural language grammar, or a stable encoding scheme. Analysis of the manuscript's word network and vocabulary evolution reveals that this assumption is untenable. The text exhibits continuous development throughout the manuscript, with vocabulary, character usage, and word frequency distributions shifting gradually from beginning to end. This dynamic character is evidenced by a single, highly connected network of similar words spanning the entire manuscript, a strong correlation between word frequency and number of similar word forms, smoothly declining cosine similarity between sections with no abrupt boundaries, and asymmetric vocabulary distribution indicating directional evolution. These properties are naturally explained by the self-citation hypothesis, which proposes that the text was generated through iterative copying and modification of previously written words—a process confirmed as the intuitive human strategy for producing meaningless text beyond approximately 100 words (Bowern & Lindemann, 2021; Gaskell & Bowern, 2022). The dynamic perspective reconciles apparently contradictory observations and provides a framework for understanding why the manuscript has resisted every analytical approach premised on static rule systems.

1. Introduction

The Voynich Manuscript, a fifteenth-century illustrated codex written in an unknown script, has defied consistent interpretation since its modern rediscovery in 1912 (D'Imperio, 1978). Despite extensive computational analysis revealing statistical patterns superficially similar to natural language—including adherence to Zipf's law, though with anomalously low conditional character entropy compared to known writing systems (Stallings, 1998; Bowern & Lindemann, 2021)—no proposed decipherment has achieved scholarly consensus (Bowern & Lindemann, 2021).

This persistent failure invites a fundamental question: is the problem that the right key has not yet been found, or that the analytical frameworks themselves are misaligned with the nature of the text? This paper argues for the latter. The common denominator of all major approaches to the Voynich Manuscript—whether seeking linguistic structures, cipher systems, or statistical regularities—is the assumption that the text was produced by a *static* system: a fixed grammar generating well-formed sentences, a stable cipher algorithm transforming plaintext into ciphertext, or a consistent encoding mapping symbols to meanings. Each approach searches for the invariant rules governing the text's production.

However, the Voynich text does not behave as the output of a static system. Its vocabulary, character usage patterns, and word frequency distributions change continuously throughout the manuscript. What appears as two discrete "languages" at a coarse level of analysis dissolves into a smooth developmental gradient at finer resolution. The text is not static but dynamic: it evolves throughout its production, and this evolution is the key to understanding why all static analytical frameworks fail.

This paper presents the evidence for the manuscript's dynamic character, demonstrates its incompatibility with static-system hypotheses, and shows how the self-citation hypothesis provides a parsimonious explanation for the observed properties.

2. The Static Text Assumption and Its Failures

Approaches to the Voynich Manuscript have differed in their specific hypotheses but converged on a shared structural assumption: the text is the product of a system whose rules remain constant throughout production. This assumption is rarely stated explicitly but pervades the analytical literature.

The cipher hypothesis assumes a fixed encryption algorithm that, once established, is applied uniformly to transform plaintext into ciphertext. Under this assumption, the same plaintext letter or word should map to the same ciphertext token (or set of tokens, in homophonic systems) throughout the manuscript. Statistical variation between sections would reflect variation in the plaintext content, not in the cipher system itself. Recent work by Greshko (2025) on the Naibbe cipher exemplifies this approach: a substitution cipher designed to replicate Voynich statistics, whose tables and transformation rules are fixed by design. Such systems, however, cannot produce the long-range correlations and continuous vocabulary evolution observed in the manuscript, precisely because their rules do not change during text production.

The natural language hypothesis assumes a grammar—whether of a known or unknown language—whose rules generate well-formed expressions. Grammatical rules do not change from page to page; vocabulary may shift with topic, but the underlying morphological and syntactic system remains stable. Researchers working within this framework have treated the statistical differences between sections as evidence of different topics or text types, not as evidence that the generative system itself is changing.

The "two languages" framework, originating with Currier (1976), illustrates how the static assumption becomes embedded in research practice. Currier observed substantial statistical differences between sections of the manuscript and labeled them "Language A" and "Language B." He regarded these differences as highly significant, warning that anyone "who attempts to work on the text without considering these, ignores them at his own peril" (Currier, 1976). Notably, however, Currier himself cautioned against over-interpreting his terminology: "Now, I'm stretching a point a bit, I'm aware; my use of the word language is convenient, but it does not have the same connotations as it would have in normal use" (Currier, 1976). Despite this explicit caveat, subsequent researchers have adopted the framework as though it described an established fact. Bowern and Lindemann (2021) write: "Following the initial observation of Currier […], we assume that there are two 'languages' in the manuscript, labeled here for convenience as Voynich A and Voynich B." This assumption of two discrete, stable systems has become a foundational premise of much subsequent analysis—precisely the kind of uncritical adoption Currier's caveat was meant to prevent. Yet as the following sections demonstrate, the manuscript's text does not partition into two discrete systems but exhibits continuous evolution from one state to the other. Currier's statistical observations remain valid and significant; it is the reification of those observations into discrete "languages" that has misled subsequent research.

The fundamental paradox that undermines all static approaches becomes apparent upon close examination. At the local level, word forms behave predictably: certain glyphs exhibit strong positional preferences (EVA-q predominantly word-initially, EVA-m predominantly word-finally), character sequences follow recognizable patterns, and morphological structures appear consistent. These regularities have led researchers to identify "rules" governing the text (Stolfi, 2003; Zattera, 2022). However, every apparent rule admits exceptions. Words like ⟨oqotar⟩ (with EVA-q in second position) and ⟨kamkam⟩ or ⟨komdy⟩ (with EVA-m in non-final positions) contradict positional "rules." More critically, when comparing different sections of the manuscript, dramatic vocabulary changes become apparent (Currier, 1976; Timm & Schinner, 2020).

The case of EVA-m illustrates the problem particularly well. Across all folios, EVA-m occurs in line-final position approximately 62% of the time—a statistic frequently cited as evidence for structural regularity. Yet this aggregate conceals dramatic local variation. In Currier A, entire bifolios contain not a single instance of EVA-m (bifolios 2/7, 10/15, 20/21, folios 25r/25v), while neighboring bifolios in the same section use it frequently (bifolio 3/6 with 27 and 12 occurrences on folios 3r and 3v respectively; bifolio 17/24 with 8 occurrences on folio 17r). Furthermore, the glyph's positional behavior shifts across the manuscript: in Currier A it occurs predominantly in word-final position, while in Currier B it appears predominantly in line-final position. What looks like a stable rule at the aggregate level is in fact a locally varying, developmentally shifting pattern—precisely the kind of behavior produced by copying from local sources whose glyph usage varies from bifolio to bifolio.

This combination—local predictability with exceptions coexisting alongside large-scale transformation—is incompatible with static rule systems. Proponents of the cipher hypothesis point to the regularities as evidence of a rule-based encoding system; advocates of the natural language hypothesis cite large-scale differences as evidence for semantic clusters reflecting different topics across sections. Neither perspective adequately accounts for both observations simultaneously: cipher systems should not exhibit such dramatic sectional variation, while natural languages should not display the degree of local regularity observed in Voynichese. The resolution lies not in choosing between these interpretations but in recognizing that the underlying assumption common to both—a static generative system—is itself incorrect.

3. The Word Network: A Single Evolving System

The most direct evidence against static-system hypotheses comes from the network structure of the Voynich vocabulary. When words are represented as nodes and edges connect words differing by a single glyph (edit distance 1), the resulting network reveals properties that are difficult to reconcile with any static generative mechanism—whether linguistic, cryptographic, or algorithmic—but emerge naturally from a dynamic copying process.

Timm and Schinner (2020) constructed this network at two scales. At the level of a single folio (f108r), the network built around the three most frequent tokens, restricted to their 33 most similar counterparts, already reveals "an existing deep correlation between frequency, similarity, and spatial vicinity of tokens within the VMS text" (p. 4). At the manuscript level, the network connects 6,796 out of 8,026 word types (84.67%), with the longest path spanning 21 steps, "substantiating its surprisingly high connectivity" (Timm & Schinner, 2020, p. 4, Figure 2).

Three properties of this network are particularly significant for the static-versus-dynamic question.

Word frequency correlates with network connectivity. The most diagnostic property of the Voynich word network—the one hardest for any competing hypothesis to explain—is the systematic correlation between a word's frequency and its number of graphically similar neighbors. Timm and Schinner (2020, p. 6, Figure 3) demonstrate that "high-frequency tokens also tend to have high numbers of similar words." The most frequent token ⟨daiin⟩ (836 occurrences) possesses 36 counterparts at edit distance 1, while isolated words—those with no similar neighbors—"usually appear just once in the entire VMS" (Timm & Schinner, 2020, p. 6). This frequency-connectivity correlation manifests in two further observations. First, the bigrams that constitute the most frequent word in a given section are also the bigrams most frequently used throughout that section—knowing which word dominates a section's frequency list effectively predicts which bigrams will dominate its bigram statistics. Second, this coupling between frequency and similarity operates at the page level: "When we look at the three most frequent words on each page, for more than half of the pages two of three will differ in only one detail. For instance, there are ten pages (f1r, f7r, f10v, f14r, f32r, f32v, f35v, f37v, f38v, and f45r) where the words ⟨daiin⟩ and ⟨dain⟩ are two out of three among the most frequent tokens" (Timm & Schinner, 2020, p. 3). This pattern is predicted by iterative copying processes, where frequent words serve as templates for further variations. It has no natural explanation under cipher or language hypotheses: in natural languages, a word's frequency is determined by its semantic and grammatical function, not by how many graphically similar words happen to exist in the text. Frequent natural language words such as articles and conjunctions do not generate families of single-edit variants clustering on the same pages.

The network is a single connected structure. The vast majority of Voynich word types belong to one network, not to separate clusters. This observation is critical because competing hypotheses predict fundamentally different network topologies. If the manuscript contained two distinct languages, two cipher systems, or text copied from multiple sources, the resulting word networks should form separate clusters—one for each language, system, or source—with few or no edges connecting them. Different languages possess different vocabularies; different cipher systems produce different output distributions; different source texts generate different word pools. The Voynich network shows none of these predicted separations. Instead, it exhibits a single, homogeneous structure in which words conventionally assigned to "Currier A" connect seamlessly to words assigned to "Currier B" through chains of minimal edits.

The network differs structurally from natural language networks. Even natural languages that superficially resemble Voynichese in certain statistical respects produce fundamentally different network structures. Timm and Schinner (2020, pp. 4–5) observe that Vietnamese, a tonal language with many similar short words, produces a connected word network when diacritical markers are omitted. However, this network shows "multiple word clusters in contrast to the homogeneous VMS-network." Furthermore, natural languages contain function words (conjunctions, articles) distributed equally across the text regardless of topic—unlike the Voynich Manuscript, where "frequently used tokens differ from page to page" (Timm & Schinner, 2020, p. 5). In natural languages, "words […] are chosen because of their meaning, and not of their similarity with previously written words" (Timm & Schinner, 2020, p. 5). The Voynich text reverses this relationship: similarity to existing words, not semantic function, appears to govern word selection.

Taken together, these network properties establish that the Voynich vocabulary was not generated by a static system—neither a fixed cipher nor a natural language grammar—but grew organically through a process in which new words emerged as modifications of existing words, with the most frequently copied words generating the most variants.

4. Continuous Evolution: The Evidence Against Discrete Systems

If the word network demonstrates that the Voynich vocabulary forms a single system rather than multiple discrete ones, the temporal dimension of that system reveals its dynamic character. The text does not merely vary between sections; it evolves continuously throughout the manuscript in a single direction, from early forms to late forms, with no abrupt transitions.

Intermediate forms and directional change. The evolution from Currier A vocabulary to Currier B vocabulary is most clearly illustrated by the progression of specific word families. The word ⟨chol⟩, highly frequent in early sections, gives way to ⟨chedy⟩ in later sections. Critically, this replacement does not occur abruptly. The intermediate forms ⟨cheol⟩, ⟨cheo⟩, and ⟨chey⟩ appear in transitional sections, documenting a gradual transformation through minimal edits: ⟨chol⟩ → ⟨cheol⟩ → ⟨cheo⟩ → ⟨chey⟩ → ⟨chedy⟩ (see Timm, 2019, section III for a detailed analysis of this shift). The existence of these intermediate forms is incompatible with discrete system changes. Spelling revision would replace one form with another cleanly; dialect switching would substitute one vocabulary for another at a boundary; copying from different source texts would produce distinct vocabularies without transitional forms. Only a process in which new words emerge as modifications of existing words—each step minimally different from its predecessor—generates such a continuous chain.

Smooth gradient without abrupt boundaries. Timm and Schinner (2024, Figure 1) provide quantitative confirmation through cosine similarity analysis between folios. The similarity declines smoothly throughout the manuscript's reconstructed writing order, with no identifiable transition point separating Currier A from Currier B. The "two languages" emerge as coarse-grained abstractions of this continuous gradient. As Timm and Schinner (2020) demonstrate, "reordering the sections with respect to the frequency of token ⟨chedy⟩ replaces the seemingly irregular mixture of two separate languages by the gradual evolution of a single system from 'state A' to 'state B'" (p. 6).

Asymmetric vocabulary distribution. The vocabulary distribution across the manuscript is asymmetric in a manner diagnostic of directional evolution. "Words typical for Currier A exist in Currier B, but not the other way round" (Timm & Schinner, 2020, p. 8). Under spelling revision or dialect switching, old forms should disappear when replaced by new ones, producing complementary rather than overlapping distributions. The observed pattern—early forms persisting alongside newly emerging forms—is consistent with accumulation through variation, not replacement through system change. Early words persist because they continue to serve as copying templates even as new variants emerge from them.

Within-section evolution. The continuous character of the evolution is further confirmed by its presence within individual sections. Zandbergen (2025) distinguishes between Stars-B (showing low correlation with Biological B) and Stars-Bio (showing high correlation with Biological B), documenting vocabulary evolution within what is conventionally treated as a single section. This within-section evolution contradicts hypotheses requiring discrete boundaries: spelling revision should not occur mid-section, dialect switching should not proceed gradually through a single text block, and a single source text cannot exhibit internal evolutionary gradients.

Feedback loops and coupled changes. The evolution is not limited to individual word families but involves coupled changes across multiple dimensions. As the text progresses, the frequency of words containing the glyph combination ⟨ed⟩ increases, words ending in ⟨y⟩ become more common, and words beginning with ⟨q⟩ proliferate. These changes are not independent: they reinforce each other through the copying process. Words ending in ⟨y⟩ statistically precede words beginning with ⟨q⟩; words beginning with ⟨q⟩ often contain ⟨ed⟩; words containing ⟨ed⟩

tend to end in ⟨y⟩. This creates self-reinforcing feedback that accelerates the adoption of certain word families—not through deliberate strategy but as an automatic consequence of copying words that exhibit these patterns.

Refutation of alternative explanations. The most common alternative explanations for the vocabulary changes—spelling revision, dialect switching, and copying from multiple source texts—share a single prediction: discrete boundaries between systems. Whether a scribe revises spelling conventions, switches from one dialect to another, or begins copying from a different source, the transition should be identifiable as a point where one system ends and another begins, with relative consistency on either side. The Voynich text shows none of this. Instead, it exhibits smooth gradients with intermediate forms, within-section evolution, and asymmetric vocabulary accumulation rather than replacement. Most critically, none of these alternatives explains why the changes should form a single directional gradient, why the word network should remain a single connected structure throughout, or why the frequency-connectivity correlation should obtain. All of these observations follow naturally from a single process of iterative copying and modification, as described in the following section.

5. The Self-Citation Process as Generative Mechanism

The network structure and continuous evolution documented in Sections 3 and 4 require a generative mechanism that produces new words as modifications of existing words, with the copying probability influenced by proximity and frequency. The self-citation hypothesis (Timm, 2014; Timm & Schinner, 2020) proposes precisely such a mechanism: the text was generated through iterative copying and modification of previously written words.

Self-citation need not be understood as a deliberate algorithm but as the natural human strategy for producing extended pseudo-text. D'Imperio (1978) observed that when faced with producing meaningless "dummy text" to conceal encrypted messages, scribes "would naturally tend to repeat parts of neighboring strings with various small changes and additions" (p. 118). Recent experimental work confirms this observation. Gaskell and Bowern (2022) asked 42 volunteers to generate meaningless text; the resulting samples were statistically similar to the Voynich Manuscript in several key respects. As Bowern and Lindemann (2021) note from a related classroom exercise: "We tested this point in an undergraduate class and found that beyond about 100 words, the task of writing language-like nonlanguage is very difficult. It is too easy to make local repetitions and words from other languages" (p. 4). As Timm and Schinner (2024) observe, this clarifies that any scribe creating language-mimicking text will sooner or later replace the tedious task of inventing more and more words by the much easier reduplication of existing text. Self-citation is therefore not a late-emerging strategy but the default cognitive approach to generating unfamiliar text.

The self-citation process naturally produces each of the properties identified in the preceding sections.

The single connected network emerges because every word is derived, directly or indirectly, from previously written words. Since the process is continuous—never interrupted by the introduction of an external vocabulary—the resulting network remains connected. New words enter the network as neighbors of existing words, ensuring that the overall structure grows organically from a single origin rather than forming disconnected clusters.

The frequency-connectivity correlation arises through a feedback loop inherent in the copying process. Frequent words are more likely to be selected as copying templates, generating more variants; the existence of more variants increases the probability that members of that word family are selected in subsequent copying events. This self-reinforcing cycle ensures that the most frequently used words accumulate the most similar neighbors—precisely the pattern documented in Figure 3 of Timm and Schinner (2020).

Continuous evolution follows from the incremental nature of the modification process. Each new word differs minimally from its source; accumulated modifications over many copying events produce gradual vocabulary drift. The direction of evolution is determined by the asymmetry of the process: new variants can only appear after their source words exist, creating a temporal arrow from early forms to late forms. "Words typical for Currier A also exist in Currier B, but not the other way round," because late-emerging variants "could not appear on pages written previously" (Timm & Schinner, 2020, pp. 8–9). This is "an automatic side-effect" of the process, not evidence of changing writing strategies.

Positional entropy and word-to-word transitions are produced by two complementary mechanisms. First, the copying process preserves structural features of source words. "The rules to modify a source word normally don't affect the order of the glyphs. This is one reason for the observation that the words in the VMS share the same rigid word structure" (Timm & Schinner, 2020, p. 10). Successful word structures become templates, propagating their positional patterns through iterative copying. Second, the scribe appears to have employed "aesthetically motivated design rules for glyph selection, in order to harmonize the overall appearance of the text" (Timm & Schinner, 2020, p. 10). Schwerdtfeger (2008) identified four such visual harmony rules governing which glyph types may follow one another. These are calligraphic preferences ensuring visual coherence, not grammatical constraints; together with structural preservation through copying, they produce the observed positional entropy patterns without requiring linguistic or cryptographic rules.

Spatial clustering of similar words results from the scribe's natural tendency to copy from nearby text. "As for the actual selection process of source words, it is clear […] that they are to be chosen at least from the same page. Since it is handy to copy a word from the same position some lines above, our implementation of the algorithm includes a mechanism that selects (with a given probability) even tokens from the previous line at the same writing position" (Timm & Schinner, 2020, p. 10). This spatial proximity in copying creates local clusters of similar words that mimic syntactic dependencies without semantic content.

Statistical tendencies rather than absolute rules are the expected output of a flexible human process. A scribe employing self-citation would develop copying habits (EVA-m usually word-final) while retaining freedom to vary (EVA-m occasionally medial). This explains why every apparent rule in the manuscript admits exceptions: they are not violations of rules but manifestations of human variability within a flexible copying process.

6. Implications of the Dynamic Perspective

Recognizing the Voynich text as the product of a dynamic process yields several implications beyond explaining the manuscript's resistance to analysis.

Reconstructing the original writing order. The continuous evolution of the word network makes it possible to recover the temporal sequence in which sections were written. The shift from Currier A to Currier B vocabulary reveals the following compositional order: Herbal (Currier A), Pharma (Currier A), Astro, Cosmo, Herbal (Currier B), Stars (Currier B), Biological (Currier B). As Timm and Schinner (2020) demonstrate, "reordering the sections with respect to the frequency of token ⟨chedy⟩ replaces the seemingly irregular mixture of two separate languages by the gradual evolution of a single system from 'state A' to 'state B'" (p. 6). The asymmetric distribution of vocabulary—words typical for Currier A exist in Currier B, but not vice versa—reveals the direction of evolution and allows reconstruction of the original compositional sequence even though the manuscript's current physical binding order differs from the textual progression.

Text-illustration correlations as temporal artifacts. The reconstructed writing order reveals a crucial pattern: with the exception of the two Herbal sections, the scribe wrote all pages sharing the same illustration type consecutively. This work organization—completing all Pharma pages together, all Astro pages together, all Biological pages together—creates the appearance of text-illustration correlation. However, the critical evidence against semantic correlation comes from the two Herbal sections themselves. Despite sharing the same illustration type (plants), Herbal A and Herbal B exhibit dramatically different vocabularies. In Herbal A, the most frequent words include ⟨daiin⟩, ⟨chol⟩, ⟨chor⟩, and ⟨shol⟩; in Herbal B, the most frequent words include ⟨or⟩, ⟨daiin⟩, ⟨aiin⟩, ⟨chedy⟩, and ⟨chdy⟩. While ⟨daiin⟩ appears in both sections (as expected for an early, persistent word), the characteristic ⟨ch⟩-initial words shift entirely: ⟨chol⟩ (228 occurrences in Herbal A) gives way to ⟨chedy⟩ (62 occurrences in Herbal B). If vocabulary tracked illustration content, both Herbal sections should exhibit similar word distributions. Instead, they differ because they were written at different evolutionary stages of the text generation process. The apparent correlations between text and illustrations reflect the coincidental alignment of section boundaries (determined by the scribe's practice of batch-processing similar illustrations) with evolutionary stages (determined by temporal progression), not semantic relationships between text content and image content. The limitations of treating these vocabulary differences as semantic topics are illustrated by the results of automated topic modeling. Sterneck, Polish, and Bowern (2021) applied three different topic modeling techniques to the Voynich text, with strikingly divergent results. As Timm and Schinner (2024) observe, "it is especially noteworthy that from the three topic modeling techniques used the Latent Dirichlet Allocation (LDA) fails completely (by allocating the entire VMS to a single topic)." The inconsistency of results across methods reflects the fundamental problem: the vocabulary differences between sections are not semantic topics but temporal snapshots of an evolving generative process.

Single authorship. The unified network evolution argues strongly for single authorship. Multiple independent authors would produce multiple independent evolutionary trajectories, resulting in distinct word networks that do not merge smoothly. The manuscript instead exhibits a single continuous development from early forms to late forms. If different sections had been written by different individuals, we would observe discontinuities in the network structure—abrupt vocabulary shifts, independent innovation patterns, or multiple coexisting evolutionary paths. The absence of such discontinuities indicates a single cognitive process unfolding over the manuscript's production, consistent with one person developing and modifying their pseudo-text generation approach over time.

The question of purpose. The question of why someone would undertake the substantial effort of creating such an artifact is separate from the question of how its text was generated. On the method, there is no mystery: self-citation is not a strategy someone consciously adopts but a cognitive default that emerges inevitably when producing extended pseudo-text. The Gaskell and Bowern (2022) experiment demonstrates this even for tasks requiring only 100 words; at the scale of the Voynich Manuscript's approximately 38,000 tokens, convergence on copying and modifying previously written words is virtually certain regardless of the scribe's original intention. As Timm and Schinner (2020) observe, "probably, the author was undergoing the substantial effort of creating the VMS in order to gain something. Not necessarily money for selling the book, although (because the algorithm can be executed by an experienced scribe almost as fast as writing down the text) the effort even for a 'classical fraud' now appears more reasonable. Perhaps it was about gaining reputation by possessing a mysterious book that no one would ever be able to decipher (simply because there really is nothing to decipher in it)" (p. 16). The illustrations—plants, astronomical diagrams, human bodies—are precisely what a medical or philosophical reference work would contain, and the impenetrable script would have discouraged anyone from questioning its content. The manuscript need not encode information; it need only look as though it does.

7. The Scale-Dependence Problem

The dynamic character of the text creates a methodological trap for researchers: the manuscript exhibits scale-dependent patterns, with different properties emerging at different levels of observation. At the character level, one finds strong positional preferences with exceptions. At the word level, local similarity networks become apparent. At the page level, substantial vocabulary fluctuation between bifolios emerges. At the section level, long-term evolutionary trends (⟨ed⟩ increase, ⟨q⟩ proliferation) dominate. At the manuscript level, the continuous evolution from Currier A to Currier B becomes visible (Currier, 1976).

Critically, what appears as a discrete boundary ("Currier A" versus "Currier B") at the manuscript level dissolves into a continuous gradient at finer scales (Timm & Schinner, 2024). The "two languages" are not objective properties of the text but coarse-grained abstractions of a continuous developmental process.

This scale-dependence means that researchers focusing on a single analytical scale will inevitably identify patterns that appear to support their hypothesis while remaining blind to contradictions visible at other scales. Those examining character-level regularities see evidence of systematic encoding; those examining section-level evolution see evidence for semantically determined topics; both are correct within their observational frame, yet neither perspective captures the complete developmental dynamics. Only analysis that spans multiple scales—recognizing that the same underlying process produces local predictability, mid-range fluctuation, and large-scale transformation—can account for the manuscript's full range of properties.

8. Methodological Implications and Conclusion

Recognition of the manuscript's dynamic, scale-dependent nature carries several methodological consequences. First, analytical approaches seeking absolute rules will inevitably encounter falsifying exceptions; the manuscript exhibits statistical tendencies reflecting copying habits, not invariant laws reflecting underlying structure. Second, discrete categorizations ("Currier A" versus "B," "regular" versus "irregular" words) are analytical conveniences, not objective properties; the text is fundamentally continuous. Third, statistical structure does not imply semantic content; the manuscript can exhibit word networks, Zipfian distributions, and topic-like clustering while remaining fundamentally meaningless, as these features emerge naturally from iterative copying processes. Fourth, traditional cryptanalytic and linguistic methods assume static systems; the manuscript requires developmental analysis, treating it as a record of an evolving process rather than an instantiation of fixed rules.

These considerations suggest that the question "Is the Voynich Manuscript meaningful?" may itself be misconceived. More productive questions include: What process generated the observed text? What constraints shaped that process? How did the process develop over time? The self-citation hypothesis provides answers compatible with all observed properties: an intuitive human copying process, constrained by cognitive limitations and aesthetic preferences, developing organically over extended production time.

The Voynich Manuscript's century-long resistance to decipherment reflects not the sophistication of its encoding but the inadequacy of analytical frameworks premised on static rule systems. The text exhibits properties of organic development: local predictability through copying habits, exceptions through human variability, and large-scale transformation through accumulated small changes. This combination—characteristic of growth processes but incompatible with rule-governed systems—explains why both cryptanalytic and linguistic approaches consistently fail.

The network structure of Voynichese, combined with experimental evidence that humans naturally employ self-citation when generating pseudo-text (Gaskell & Bowern, 2022), suggests a parsimonious explanation: the manuscript represents the trace of a human cognitive process unfolding over time, generating meaningless text through intuitive copying and modification. This process naturally produces the observed combination of language-like statistics and systematic peculiarities without requiring either sophisticated encryption or genuine semantic content.

It should be acknowledged that the self-citation algorithm as implemented by Timm and Schinner (2020) does not reproduce every statistical subtlety of the Voynich text. However, this does not disprove the underlying concept. As Timm and Schinner (2024) observe: "The algorithm outlined in Timm & Schinner 2020 does not reproduce some of the more subtle statistical observations (as several of our critics have emphasized), but this does not disprove the underlying concept. […] The VMS was created by a human being with all freedom to spontaneously make and break (intuitive or explicit) own rules. After all, from the viewpoint of complexity theory, most likely the minimal representation of the VMS is the manuscript itself" (p. 319).

Future research should focus less on seeking the "key" to the manuscript and more on characterizing the developmental process that generated it. Understanding the Voynich Manuscript requires treating it not as a puzzle with a solution but as a dynamic artifact recording an extended episode of human text production—one that inadvertently created patterns regular enough to appear systematic yet flexible enough to evade every systematic explanation.

References

Bowern, C. L., and Lindemann, L. (2021). The linguistics of the Voynich manuscript. Annual Review of Linguistics, 7, 285–308.

Currier, P. H. (1976). New research on the Voynich Manuscript. In M. E. D'Imperio (Ed.), Proceedings of a Seminar. Washington, DC: Privately Printed Pamphlet.

D'Imperio, M. E. (1978). The Voynich Manuscript: An elegant enigma. Laguna Hills, CA: Aegean Park Press.

Gaskell, D. E., and Bowern, C. L. (2022). Gibberish after all? Voynichese is statistically similar to human-produced samples of meaningless text. International Conference on the Voynich Manuscript 2022, University of Malta. https://ceur-ws.org/Vol-3313/paper4.pdf

Greshko, M. A. (2025). The Naibbe cipher: a substitution cipher that encrypts Latin and Italian as Voynich Manuscript-like ciphertext. Cryptologia, 1–37.

Schwerdtfeger, E. (2008). Die Harmonie der Glyphenfolgen. Retrieved from http://voynich.tamagothi.de/2008/07/05/die-harmonie-der-glyphenfolgen/

Stallings, D. (1998). Understanding the second-order entropies of Voynich text. Ixoloxi.com, May 11. http://ixoloxi.com/voynich/mbpaper.htm

Sterneck, R., Polish, A., and Bowern, C. (2021). Topic Modeling in the Voynich Manuscript. arXiv:2107.02858 [cs.CL]. doi: 10.48550/arXiv.2107.02858

Stolfi, J. (2003). Voynich manuscript stuff. Retrieved from http://www.ic.unicamp.br/~stolfi/voynich/

Timm, T. (2014). How the Voynich Manuscript was created. arXiv:1407.6639.

Timm, T. (2016). Co-Occurrence Patterns in the Voynich Manuscript. arXiv:1601.07435.

Timm, T. (2019). Shift from <chol>/<chor> to <chedy>/<shedy>. https://github.com/TorstenTimm/SelfCitationTextgenerator/blob/master/chey_chedy.md

Timm, T., and Schinner, A. (2020). A possible generating algorithm of the Voynich manuscript. Cryptologia, 44(1), 1–19.

Timm, T. and Schinner, A. 2021. Review of The linguistics of the Voynich manuscript by Claire Bowern and Luke Lindemann. Cryptologia 45 (5): 434-438.

Timm, T., and Schinner, A. (2024). The Voynich manuscript: Discussion of text creation hypotheses. Cryptologia, 47(4), 307–322.

Zandbergen, R. (2025). Currier A and B: two different languages. https://www.voynich.nu/extra/lang.html (accessed February 21, 2026).

Zattera, M. (2022). A new transliteration alphabet brings new evidence of word structure and multiple ‘languages’ in the Voynich Manuscript. International Conference on the Voynich Manuscript 2022, University of Malta.